

A Novel Gaze Gesture Sensor for Smart Glasses Based on Laser Self-Mixing

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ABSTRACT

The integration of gaze gesture sensors in next-generation smart glasses will improve usability and enable new interaction concepts. However, consumer smart glasses place additional requirements to gaze gesture sensors, such as a low power consumption, high integration capability and robustness to ambient illumination. We propose a novel gaze gesture sensor based on laser feedback interferometry (LFI), which is capable to measure the rotational velocity of the eye as well as the sensor's distance towards the eye. This sensor delivers a unique and novel set of features with an outstanding sample rate allowing to not only predict a gaze gesture but also to anticipate it. To take full advantage of the unique sensor features and the high sampling rate, we propose a novel gaze symbol classification algorithm based on single sample. At a mean F1-score of 93.44 %, our algorithms shows exceptional classification performance.

KEYWORDS

gaze interaction, laser feedback interferometry (LFI) sensor, smart glasses interaction

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1 INTRODUCTION

In recent years, various smart glasses have been released into the market. As a successor of virtual reality (VR) glasses, they follow a more natural design while integrating similar sensing technology into the glasses. In general, they consist of a light projection engine and a set of sensors to capture user inputs and monitor the user's state.

To enable human computer interaction (HCI) with smart glasses, a variety of interaction concepts are possible. These concepts can be classified into touch based interactions e.g. via a track pad on the glasses temple introduced by the Google Glass [13], an external on-body device e.g. a controller like in the Magic Leap One [17] or via touchless interactions such as spoken commands via voice recognition as used by the Echo Frames [1]. Also, hand gestures captured by a camera sensor have been shown by the HoloLens [12].

An additional touchless interaction concept is making use of the user's gaze by tracking their eye movements by means of eye-tracking sensors. Gaze-based interaction allows a fast and natural input, leading to an intuitive and unobtrusive way of interaction with smart glasses while maintaining social acceptance and user's privacy [12].

To the best of our knowledge, no commercially available smart glasses solution so far utilizes gaze gestures as input modality, mainly due to the challenges arising from integration and due to limited available power and space constraints [12]. Furthermore, a high robustness of the sensor against variable external illumination is required to allow for operation in uncontrolled outdoor environments e.g. in bright sun light [6].

To overcome these limitations and enable gaze gesture based interaction for the next generation of smart glasses, we present a novel low power gaze gesture sensing approach based on laser feedback interferometry (LFI). This multimodal sensor is capable of measuring distance towards the eye as well as eye rotational velocity with a high sample rate of up to 1 kHz. The sensor is based on a small vertical cavity surface emitting laser (VCSEL) with a cavity-integrated photodiode in the near infrared (IR) spectrum, which allows for a space constrained integration into the glasse's frames.

The VCSEL with cavity-integrated photodiode enables to apply the LFI sensing principle, a self-coherent sensing scheme. This leads to a high sensitivity to its own radiation while suppressing most radiation entering the laser cavity that is not their own. This allows a robust operation in presence of ambient radiation, as shown in previous studies [16].

The main contributions of this work are the introduction of the novel LFI-based sensing principle for low-power gaze gesture symbol recognition and a single sample classification approach optimised for the feature set obtained from the LFI sensors.

In the next section, we give an overview of the state of the art regarding gaze gesture sensor concepts for AR glasses and discuss the limitations of existing sensor concepts. In Section 3, we introduce the sensing principle and the sensor concept. In addition, we provide an overview of the measurement features produced by the LFI sensor on the human eye. Based on these features, we introduce an optimized and robust gaze symbol classification algorithm in Section 4. Afterwards, we evaluate the proposed sensor concept in Section 5 using a laboratory setup. In the last section, we conclude our findings and discuss further steps.

2 RELATED WORK

Several eye-tracking sensor concepts have been investigated in the past. The successors of these different concepts, which are widely used in research and commercial applications, are video-oculography (VOG) and electro-oculography (EOG) [14]. In addition, novel low power eye-tracking sensor approaches for AR glasses based on microscanners and infrared (IR) lasers have emerged in recent years [15, 20].

Based on VOG and EOG, several gaze-based interaction concepts have been developed. Early work attempts to use gaze as a pointing device to interact with UI elements by dwelling on them [2]. The main drawbacks are a small number of interactions in a given time period and a required absolute calibration of the eye tracker.

To overcome this limitations Drews et al. introduced a gaze gesture based approach. They used a VOG eye-tracking sensor with a 60 Hz sample rate to capture the absolute gaze position and extract gaze gestures from the input data stream [9]. They described a gaze gesture as a sequence of atomic eye movements. Atomic eye movements, also refereed to as strokes of the eye, are single unidirectional eye movements, e.g. left or upwards and can be interpreted as gaze symbols. A set of gaze symbols forms a gaze gesture protocol. Drews et al. used gaze symbols to control the user interface by linking a sequence of gaze symbols to a gaze gesture.

The main disadvantage of VOG sensor based approaches is the high power consumption required by the sensors and the image processing, as well as the limited sample rate of the sensors [12]. Furthermore, they are sensitive to ambient light which disturbs the captured images leading to a low detection rate and therefore limiting the interaction capabilities in the wild [6].

To overcome this limitations, Bulling et al. integrated EOG sensors into a wearable glasses demonstrator to measure relative movements of the eye and extract gaze gestures for mobile human computer interactions [4]. They placed two electrode pairs for horizontal and vertical eye movements around the eye and used a microcontroller to sample the four EOG channels and extract the

relative eye movements. In addition, they used a light sensor and an accelerometer to compensate external artefacts in the sensor data. They reported a sample rate of 250 Hz with an overall power consumption of 769 mW.

They use a gaze gesture protocol with 16 symbols to encode the gaze gestures. The overall correct classification rate for different gaze gestures of varying complexity was given as 87 %.

Compared to VOG sensor concepts, EOG sensors benefit from computational light weight signal processing. In addition, they are more robust against ambient illumination. The major drawback limiting the use of EOG sensors for gaze-based interaction in recent smart glasses is the use of electrodes on the skin [14].

Our gaze gesture sensor concept based on LFI combines the benefits of non-intrusive integration of VOG sensors and the calibration free measurement of relative eye movements by EOG sensors. Furthermore, our sensor principle is robust against external illumination and is therefore capable of operating in the wild.

3 GAZE GESTURE SENSOR BASED ON LASER FEEDBACK INTERFEROMETRY

Laser feedback interferometry or self-mixing interferometry (SMI) is a widely known interferometry measurement method. It is used in the industry as well as in laboratory environments to measure displacement and velocity of solid targets, as well as fluids and distance [7].

3.1 Sensing principle

The sensing principle is described by the coupled cavity model, shown in Figure 1 b) shows the coupled cavity model. A laser with an optical output power P_0 emits a coherent laser beam towards the surface of the eye. The laser beam hits the eye under an angle of incidence γ , is attenuated by volume scattering effects and absorption described by a reflectivity R and is back injected into the laser where it causes stimulated emission of the laser, just as the oscillating photons in the laser cavity do. However, as it took the external light a time τ_{ext} to travel the distance L_{ext} towards the eye, the laser field inside of the cavity and the back-injected laser field might be in phase (constructive interference, increase in laser field strength) or out of phase (destructive interference, decrease of laser field strength).

The resulting modulated optical output power by self-mixing interference is dependent on the unmodulated nominal output power P_0 , the modulation depth m and a varying phase ϕ_{fb} of the backscattered light field

$$P'_0 = P_0 \left(1 + m \cdot \cos(\phi_{fb}) \right). \quad (1)$$

In our sensor, a small fraction of the modulated output power P'_0 is then measured by a photodiode, which is integrated into the distributed Bragg reflector structure of the laser cavity itself [22].

To understand the link of the phase ϕ_{fb} to our observation goal (distance and velocity), we consider the rate equations introduced by [11]. Here, a solution of the rate equations is the excess phase equation which is only dependent on the signal phase ϕ_s in our case (so called weak feedback regime) [22], leading to

$$\phi_s = \frac{4\pi n_{ext} L_{ext}}{\lambda} \quad (2)$$

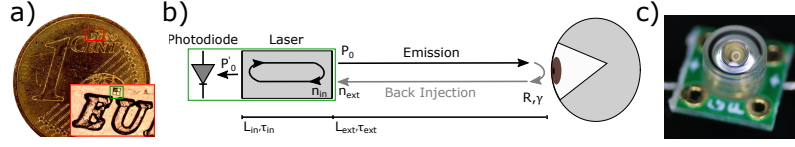


Figure 1: a) 160 μm x 180 μm VCSEL with integrated photodiode on a cent coin b) Coupled cavity model of a laser feedback interferometry sensor. The laser emits light which is scattered by the eye and back injected into the laser cavity. The photodiode monitors the laser power, which varies with changes in the feedback path. c) Encapsulated optical module including the LFI sensor and the beam shaping optics.

with λ describing the wavelength of the laser. Considering $n_{ext} \approx 1$ is constant due to the operation of the sensor in free space, only changes in λ and L_{ext} lead to a varying phase ϕ_s and, in consequence, to a varying phase ϕ_{fb} . This results, with respect to Equation (1), in an oscillating modulation of the optical output power. The frequency of oscillation observed by the photodiode can be considered the beat frequency of optical mixing in the laser cavity. Mapping of the beat frequency to velocity and distance is analogous to frequency modulated continuous wave (FMCW) lidar systems.

While the beat frequency of a moving object appears also for continuous wave operation of the laser, distance measurements require a modulation of the laser wavelength [3]. One option to introduce the required wavelength shift is modulating the laser drive current. Due to finite efficiency, the laser temperature follows the modulation current pattern and so do cavity length and lasing wavelength. Both contributions to the beat frequency (velocity and distance) can be considered individually by the partial derivative of Equation (2) with respect to time, which leads to

$$f_0 = \frac{2L_{ext}}{\lambda^2} \frac{d\lambda}{dI} \frac{dI}{dt} \quad L_{ext}=const. \quad (3)$$

and

$$f_d = \frac{2v_{ext} \cos(\gamma)}{\lambda} \quad \lambda=const. \quad (4)$$

Movements of the eye ($\frac{dL_{ext}}{dt} = v_{ext}$) lead to a shift of the beat frequency f_0 by a Doppler frequency component f_d . If the angle of incidence γ is known, Equation (4) can be rearranged with respect to v_{ext} to obtain the surface velocity of the eye.

Therefore, without incorporating further a-priori assumptions two LFI sensors are required to measure both, the horizontal movement around the θ -axis and vertical movement around the ϕ -axis. Figure 2 shows an exemplary positioning and the laser beam directions of two LFI sensors to comply with these requirements.

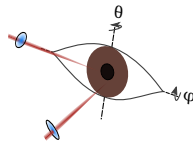


Figure 2: Positioning of the LFI sensors with respect to the rotational axes of the human eye.

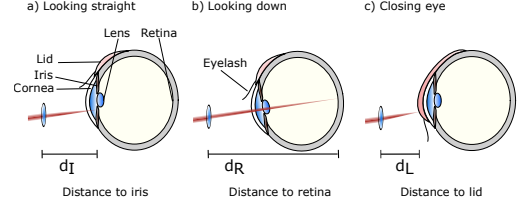


Figure 3: Different movement related position changes of the eye and the lid resulting in a change in distance measurement due to the geometry and anatomy of the eye.

From an integration point of view, the sensor for vertical movements can be integrated into the AR glasses frame below the spectacle and the sensor for horizontal movements can be integrated into the AR glasses frame temple. The size of a single LFI sensor is mainly determined by the diameter of the lens in front of the laser chip as shown in Figure 1 c). The diameter of the lens is in the range of 1 mm - 2 mm.

3.2 Sensor features on the eye

Based on Equation (3) and Equation (4) and the positioning of the LFI sensors shown in Figure 2, four features are measured by the sensors. For each rotational axis of the eye, the surface velocity and the distance are measured resulting in v_θ and v_ϕ as velocity related features and d_θ and d_ϕ as position related features.

Figure 3 shows a sectional view of the eye and a fixed LFI sensor position for three different positions of the eye and the lid. In Figure 3 a), the gaze is directed straight ahead. The laser beam of the LFI sensor for vertical rotations penetrates the cornea and backscattering occurs at the iris. For this setup, the LFI sensor measures the distance between sensor and iris d_I .

In Figure 3 b), the eye is slightly rotated along the vertical direction downwards and the laser beam penetrates the cornea and the lens and is backscattered from the retina. The measured distance in this case is the distance between sensor and retina d_R .

In Figure 3 c), the eye is directed straight forward but the lid is closed by a blink. In this case, the distance d_L between sensor and lid is measured. By subtracting d_I and d_L , the thickness of the lid and the cornea can be calculated, which is around 4 mm, and by subtracting d_R and d_L the diameter of the eye is approximately calculated, which is around 24 mm [8].

4 GAZE GESTURE ALGORITHM

Similar to related works by [9], [4] and [5], the gaze gesture algorithm is based on a gaze gesture protocol which encodes atomic movements of the eye into gaze gestures denoted by symbols.

We define a gaze symbol set consisting of four basic atomic eye movements which are annotated by small letters. For the horizontal axis, movements to the left l and to the right r around θ are possible. In addition, eye movements in the vertical axis around ϕ are denoted by u and d for up and down movements of the eye, respectively. In addition to movements of the eye, blinks (b) are considered an additional atomic movement. In order to cover all types of eye movements, fixations of the eye as well as slow movements are described by an additional symbol n for non-movements.

The entire set of symbols S thus consists of six unique symbols $S = \{l, r, u, d, b, n\}$.

4.1 Feature extraction

The novel sensor concept described in Section 3 allows for the extraction of a unique set of features from the human eye with a high sampling rate. A major advantage over the state of the art is the use of the distance measurement between the glasses' frame and the eye as an additional feature, as discussed in Section 3.2.

To obtain unique features from the measured velocities, v_θ and v_ϕ are treated as velocity vector components. This velocity vector can be represented in polar coordinate space by an angle

$$\epsilon = \arctan\left(\frac{v_\theta}{v_\phi}\right) \quad (5)$$

and a vector length

$$v = \sqrt{v_\theta^2 + v_\phi^2} \quad (6)$$

which leads us to a two dimensional feature vector F with ϵ and v as gaze symbol features. Figure 4 shows the feature space, which is spanned by ϵ and v , in polar coordinate space. The grey area in the centre covers the area of fixations and slow eye movements e.g. drift or tremor. The size is determined by a certain absolute velocity threshold v_n for v . The sensitivity and robustness of the gaze symbol algorithm can be controlled by varying v_n .

If the glasses slip on the head, the sensor positions and thus the intersection points between the laser beams and the eye are affected. To be robust against the translational and rotational motions of the sensors with respect to the eye introduced by glasses slippage, the classifier allows a wide range of angles of 90° per direction. This improves the robustness of the gaze symbol detection and allows an easy execution of gaze gestures for the user.

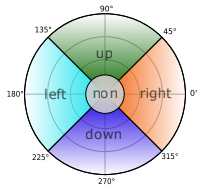


Figure 4: Gaze gesture velocity feature space shown in a polar coordinate system.

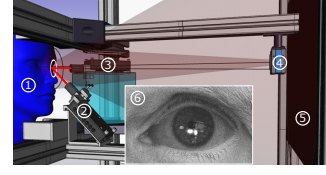


Figure 5: CAD sketch of the laboratory setup to validate the proposed LFI gaze gesture sensor approach.

To also be able to measure blinks d_θ is added as a third feature to F . The feature vector describes a point in cylindrical coordinates denoted by $F = \{\epsilon, v, d_\theta\}$.

4.2 Gaze symbol classification

To classify the gaze symbols a classifier maps a feature vector F measured by the LFI sensors to a gaze symbol from the gaze symbol set S . For this purpose, a decision tree, which describes the feature space shown in Figure 4 and the additional distance information d_θ , is used as classifier. The main advantage of this single-sample classification approach over other classification approaches, e.g. by [4], is the invariance with respect to time. This allows for a robust classification, which is insensitive to sensor drift, as is the case with EOG sensors [14]. Furthermore, this classification approach does not require the detection of a movement sequence in the input sensor signal stream to extract significant movements which belong to a gaze symbol as proposed by [5].

5 EVALUATION

To validate the proposed features of the human eye measured by the LFI sensors and evaluate the gaze gesture recognition algorithm, a laboratory setup is used, as shown in Figure 5.

The subject ① is placed in front of a monitor ⑤ which is used to create stimuli for eye movements. The LFI sensor at position ② is responsible for measurements of v_ϕ and d_ϕ while the LFI sensor at position ③ is responsible for measurements of v_θ and d_θ . An IR camera ④ captures a video of the subject during the experiments. ⑥ shows a captured video frame showing the laser spots on the iris of a subject. The laser is current modulated by an triangle signal generated by an waveform generator and the photodiode current of both LFI sensor is captured by an oscilloscope. The IR VCSEL and the driver electronics as well as the analogue signal processing are fabricated in house.

Table 1: Properties of the LFI sensors of the laboratory setup.

LFI Sensor	$d\lambda/dI$	λ	γ	P_0
②	0.406 nm/mA	848 nm	45°	$390 \mu W$
③	0.396 nm/mA	856 nm	45°	$390 \mu W$

With the low optical power of the IR VCSELs, the mechanical setup of chin and head rest and the positioning of the lasers in the laboratory setup, a class 1 laser system according to IEC 60825-1 (well below class 1 optical power limit of $770 \mu W$) is achieved and, therefore, the experiments do not pose any medical hazard to the subject's eye.

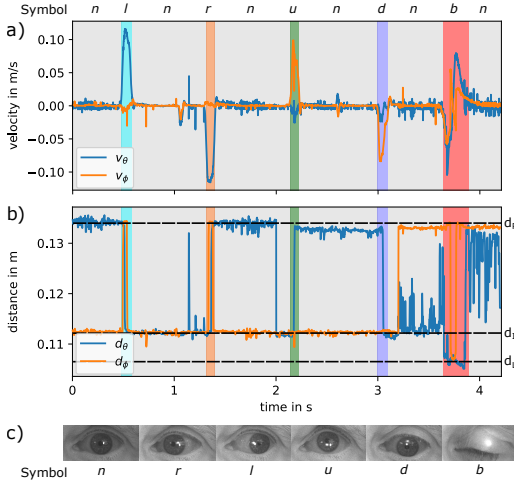


Figure 6: Data captured with the laboratory setup to evaluate the proposed features of the eye. a) shows the measured velocity, b) shows the measured distance and c) shows eye movements captured with the IR camera and the corresponding gaze symbol. The background colour refers to the corresponding atomic movement of the eye.

The power consumption of the proposed gaze gesture sensor is roughly 140 mW, which is estimated based on an STM32G473 microcontroller including all required peripherals to capture and process the data from the LFI sensors. With optimized logic blocks and a subsampling scheme enabled by a higher integration using a custom application specific integrated circuit (ASIC), a further power reductions to 30 mW is expected. A comparable power consumption of 150 mW is reported by [20] for state of the art VOG eye tracking sensors, however excluding image processing. Compared to the EOG approach of Bulling et al. [4], a much lower power consumption can be achieved.

5.1 Feature validation

To validate the proposed features of the human eye from Section 3.2, the laboratory setup is used. Figure 6 shows the measured data of the two LFI sensors from a subject performing the gaze symbol sequence $[l, r, u, d, b]$. The LFI sensors are modulated to achieve an update rate of 1 kHz for distance and velocity measurements. Figure 6 a) shows the measured velocity and Figure 6 b) shows the measured distance. The movement threshold v_n is set to 0.02 m/s to distinguish between atomic eye movements and non-relevant movements of the eye. In the distance plot we added three distance lines corresponding to the distances described by Figure 3 a) - c). The measured distance between d_R (retina) and d_I (iris) is 23,36 mm and the measured distance between d_I (iris) and d_L (lid) is 5,01 mm, which both correspond to the known anatomy of the human eye.

5.2 Gaze gesture symbol classification

To validate the proposed gaze gesture symbol classifier described in Section 4.2, a set of gestures is recorded using the laboratory setup as described in Section 5. Two male subjects with blue and brown

eyes were instructed to perform 18 eye movement gestures and 9 blinks each. To capture natural trajectories of the eye, during the execution of the gaze gestures no visual stimuli were used to guide the gaze of the subjects.

In a first step, the ground truth is manually annotated utilizing the IR camera images to the measured data with the corresponding gaze symbols, similar to related work by [19]. To distinguish between movements (saccades) and non-movements (fixations), a velocity threshold $v_n = 0.02$ m/s is applied. Considering Emsley's reduced eye model ($r_{eye} = 11.11$ mm) this equals to a velocity threshold of $103.14^\circ/\text{s}$, close to the suggested velocity threshold used in the velocity threshold Identification algorithm (I-VT) [18].

Afterwards, we extract the features from the measured data according to Section 4.1 and use the proposed gaze symbol classification algorithm to classify atomic eye movements. This allows us to evaluate the classification approach on the single sample-level. In addition, we treat the multiclass classification problem for evaluation purposes as a binary one-vs-all classification problem, which allows to use the F1-score as evaluation metric [10, 21]. Table 2 shows the evaluation results.

Table 2: Evaluation results of the decision tree based gaze symbol classifier based on a single sample base.

Gaze symbol	Gestures	Symbols	Precision	Recall	F1-Score
<i>l</i>	36	3783	0.906	0.995	0.949
<i>r</i>	36	3505	0.916	0.997	0.955
<i>u</i>	36	3229	0.820	0.976	0.891
<i>d</i>	36	3624	0.873	0.945	0.908
<i>b</i>	18	3063	0.971	0.967	0.969

6 CONCLUSION

We present a novel low power gaze gesture sensor concept based on IR VCSELs with an integrated photodiode which show the LFI effect. Their extraordinary small size allows for seamless integration of gaze gesture sensors into next generation smart glasses and enables new gaze based interaction concepts like true hands-free interaction. We also introduced a computational lightweight gaze symbol classification algorithm with a sufficient classification accuracy and precision.

Based on the promising results obtained with the laboratory setup, our next goal is to integrate the sensors into our head-mounted demonstrator. Furthermore, we want to develop a gaze gesture algorithm which recognizes a user interaction with the glasses by detecting a sequence of gaze symbols and investigate the robustness against unwanted interactions during everyday activities.

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